Simulation Study: Implementation

Boschi, Martina

2026-04-01

Synthetic Data: Academics moving in US states.

Preliminarities

More details about the preliminaries can be found in the document: `01-simulation-study-implementation`.

```
load(file="output/remdats_data.RData")
load(file="output/gam.fits.RData")
load(file="output/shifted_rems.RData")
load(file="output/gam.fits_shifted.RData")
```

```
source("script/states.R")
```

```
##   |                                                                      |
```

```
source("script/input_distance.R")
source("script/true_coefficients.R")
```

```
n_sim = 100
```

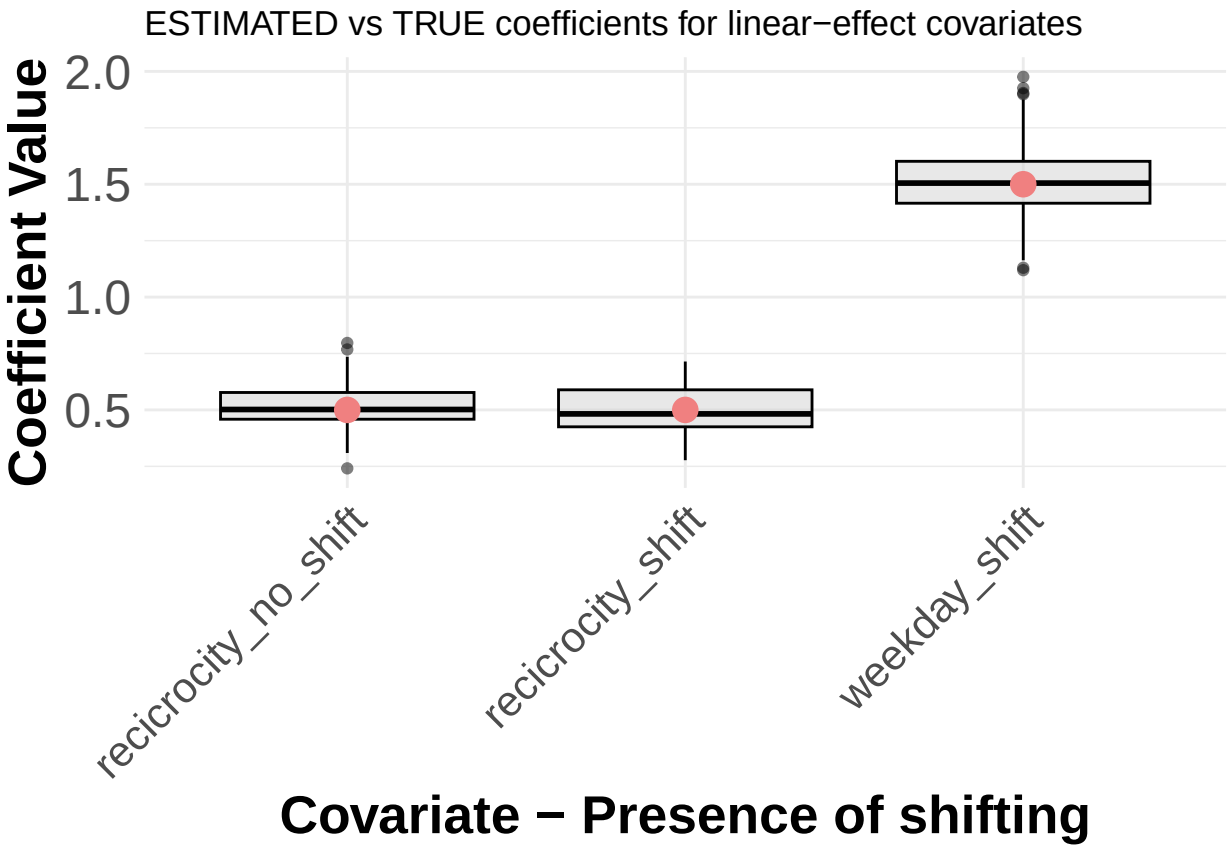
Fixed Effects

```
# Create the data frame as you did
fixed_results <- data.frame(
  recicrocity_no_shift = unlist(lapply(gam.fits, function(x) x$coefficients[1])),
  recicrocity_shift = unlist(lapply(gam.fits_shifted, function(x) x$coefficients[1])),
  weekday_shift = unlist(lapply(gam.fits_shifted, function(x) x$coefficients[2]))
)

# Reshape data to long format for ggplot
fixed_results_long <- fixed_results %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")

coefficients_fixed <- ggplot(fixed_results_long, aes(x = Variable, y = Value)) +
  geom_boxplot(aes(fill = Variable), color = "black", alpha = 0.5) +  # Boxplot with color
  geom_point(data = data.frame(Variable = c("recicrocity_no_shift",
                                            "recicrocity_shift",
                                            "weekday_shift"),
                               Value = c(b1, b1, b0)),
             aes(x = Variable, y = Value),
             color = "lightcoral", size = 4, shape = 19) +  # Add points
  labs(title = "ESTIMATED vs TRUE coefficients for linear-effect covariates",
       x = "Covariate - Presence of shifting",
       y = "Coefficient Value") +
```

```
  scale_fill_manual(values = c("recicrocity_no_shift" = "lightgray",
                               "recicrocity_shift" = "lightgray",
                               "weekday_shift" = "lightgray")) +  # Custom colors
  theme_minimal() +
  theme(legend.position = "none",  # Remove legend as it's not needed
        axis.text.x = element_text(angle = 45, hjust = 1, size = 16),
        axis.text.y = element_text(size = 18),
        axis.title.x = element_text(size = 20, face = "bold"),
        axis.title.y = element_text(size = 20, face = "bold"),
        )
coefficients_fixed
```



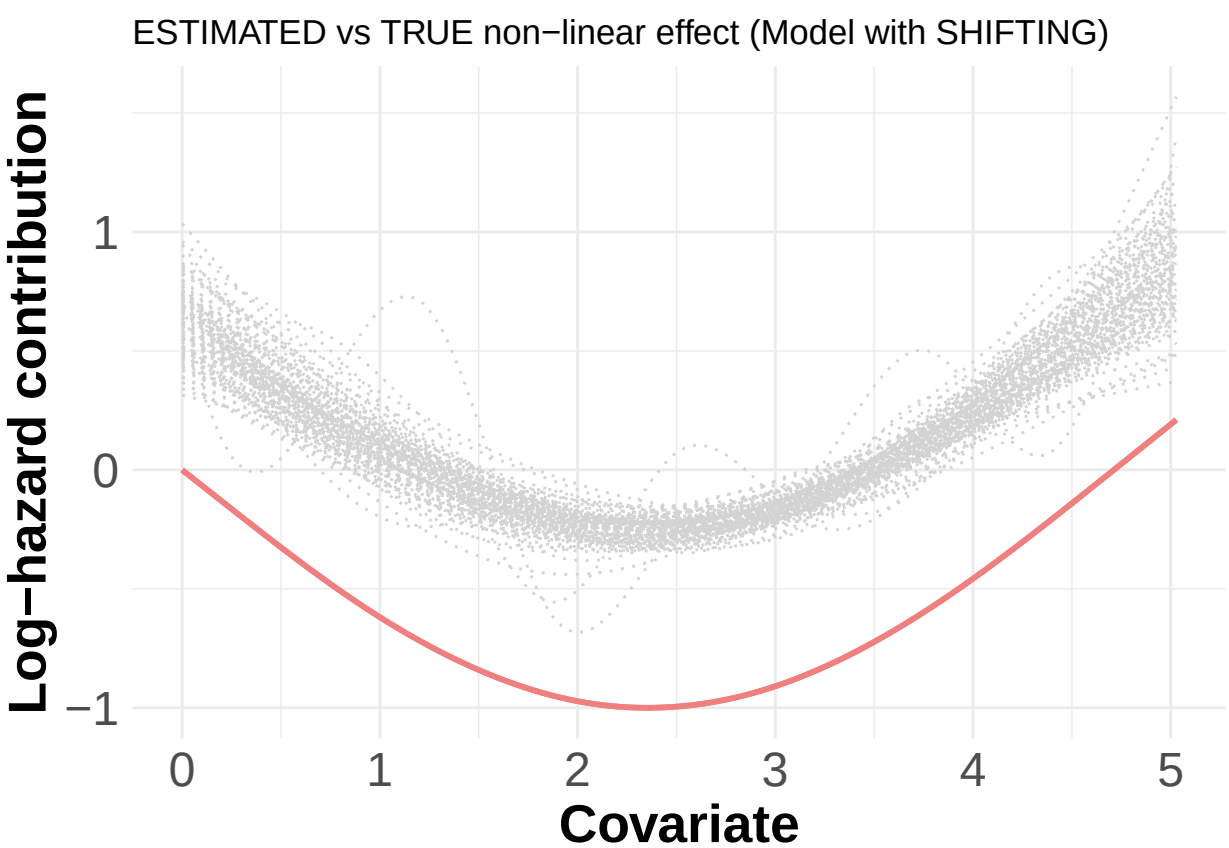

```
pdf("pictures/simulation/estimates_simulation_complete.pdf", width=10, height=7)
coefficients_fixed
dev.off()
```

```
## pdf
##   2
```

Non-linear effect

```
b.dist_shift.objects <- lapply(gam.fits_shifted, getViz)
data <- data.frame(x = as.vector(dist_matrix),
                   y = spline_function(as.vector(dist_matrix), fit))
coefficients_plot_shifted <- ggplot(data=data, aes(x, y)) +
  geom_line(data=data,
            linetype = 1, col="lightcoral",
```

```r
            size=1)
for (iter in 1:n_sim){
  b.iter <- b.dist_shift.objects[[iter]]
  o.iter <- plot(sm(b.iter, 1))
  coefficients_plot_shifted <- coefficients_plot_shifted +
    geom_line(data=o.iter$data$fit,
              linetype = 3, col="lightgray",
              size=0.5)
}
coefficients_plot_shifted <- coefficients_plot_shifted +
  geom_line(data=data,
            linetype = 1, col="lightcoral",
            size=1) +
  labs(title = "ESTIMATED vs TRUE non-linear effect (Model with SHIFTING)",
       x = "Covariate",
       y = "Log-hazard contribution") +
  theme_minimal() +
  theme(legend.position = "none",  # Remove legend as it's not needed
        axis.text.x = element_text(size = 18),
        axis.text.y = element_text(size = 18),
        axis.title.x = element_text(size = 20, face = "bold"),
        axis.title.y = element_text(size = 20, face = "bold"),
        )
coefficients_plot_shifted
```




```r
pdf("pictures/simulation/non-linear-effect-simulation.pdf", width = 10, height = 7)
coefficients_plot_shifted
```

```r
dev.off()
```

```
## pdf
##   2
```